**TALLINN UNIVERSITY OF TECHNOLOGY**

# MES0225 MACHINE ENGINEERING – PROJECT

# ELECTRIC WINCH

Design Report


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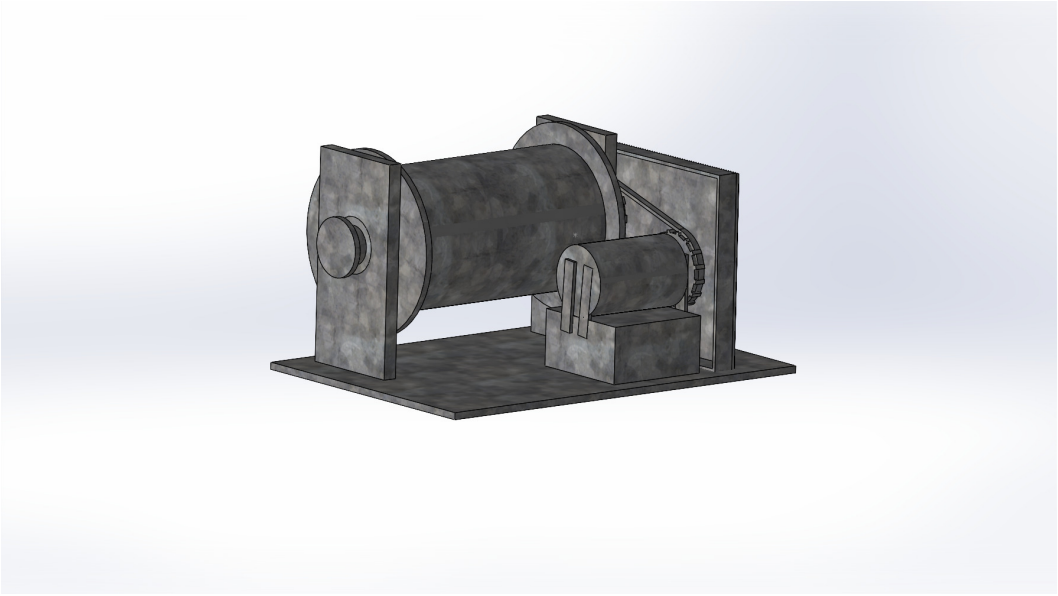



Figure 1: Assembly model of the electric winch.

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# 1 Initial Conditions

## 1.1 Name

Electric winch (EST. *Elektriajamiga trummelvints*, GER. *Elektrische Winde*, RUS. ).

## 1.2 Purpose and Field of Use

An electric winch is a motorised traction device designed to lift, lower, and move loads by winding a rope around a powered drum. The designed device is intended for lifting and positioning heavy components in an assembly workshop, where manual handling would be impractical or unsafe.

The winch can be used independently or together with overhead structures and similar material-handling systems. Typical areas of use include manufacturing facilities, warehouses, and industrial workshop environments.

## 1.3 Basic Functions, Parameters and Technical Requirements

- load capacity: 750 kg,
- rope velocity: 0.11 m/s,
- connection between gear motor and drum: chain drive,
- during the design, minimise dimensions and mass of the device.

Base functions:

- lifting of heavy-weight components in assembling shop,
- simplification of assembly operations.

As a prime mover, a spur- or worm gear motor is used, connected to the drum by means of a chain drive.

## 1.4 Base Materials

The drum is welded from steel components. Drum material: S235J2G3 EN 10025. The drum is supported on the shaft by means of two hub units. Shaft material: C45E EN 10083. Torque is transmitted from the drum to the shaft by keys located in both hub units. The shaft is supported on self-aligning bearings. Bearing units are fastened to the frame by bolts. The frame is fabricated from steel pipes (S355J2H) or UNP sections (S235JRG2).

## 1.5 Mechanical Properties of Materials

Table 1: Mechanical properties of materials

| Property | Steel S235 | Steel S355 | Steel C45E |
|---|---|---|---|
| Yield stress $R_{eH}$ / $\sigma_Y$ | 235 MPa | 355 MPa | 370 MPa |
| Ultimate stress $R_m$ / $\sigma_U$ | 370–470 MPa | 490–610 MPa | 630 MPa |
| Fatigue limit $\sigma_{-1}$ | – | – | 275 MPa |
| Fatigue limit $\tau_{-1}$ | – | – | 165 MPa |

Modulus of elasticity: $E = 2.1 \times 10^5$ MPa
Shear modulus: $G = 8.1 \times 10^4$ MPa

### 1.6 Compliance with Special Conditions

- Reliability – designed for a minimum of 500 full-load operating cycles before scheduled maintenance; sealed bearings and thermal motor protection ensure stable long-term operation.
- Environmentally friendly – electric drive produces zero direct exhaust emissions; lubricants must not enter the environment; target noise level $\leq$ 72 dB(A) at 1 m under rated load; RoHS-compliant materials.
- Safety – once a year the rope is tested for strength; automatic load-holding brake (fail-safe, engages on power loss); overload cut-off at 110–120% of rated load; compliance with Machinery Directive 2006/42/EC.
- Climate resistance – operating temperature -10°C to +40°C; sealed motor housing IP55; galvanised or powder-coated drum and frame.
- Aesthetics and ergonomics – compact rectangular housing; large clearly labelled IN/OUT buttons on pendant controller; freespool lever accessible without tools.
- Quality – production quality per ISO 9001; rope tested to EN 12385; mechanical components proof-loaded to 1.5× rated load; full functional test before dispatch.

## 2 Kinematic Diagram

The kinematic diagram shows the power transmission path of the device. The gear motor is mounted on the frame and transmits torque to the drum via a chain drive. The drum winds the rope to lift and lower the load.

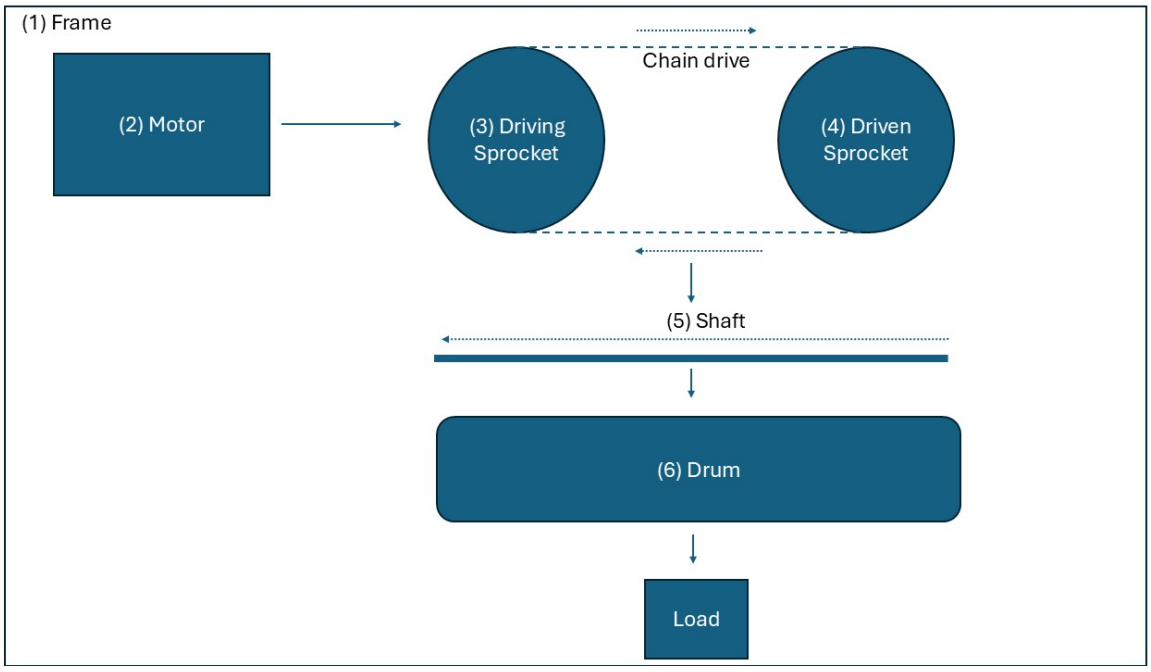



Figure 2: Kinematic diagram of the electric winch showing power transmission from the motor to the drum via a chain drive.

Figure description: 1 – frame, 2 – gear motor, 3 – chain drive, 4 – drum.

# 3 Rope Selection and Calculation of Drum Diameter

The maximum internal force of the rope must satisfy:

$$F_{\max} \leq [F] = \frac{F_{kr}}{S} \tag{1}$$

The maximum tensioning force:

$$F_{\max} = mg = 750 \times 9.81 \approx 7360\,\text{N} \tag{2}$$

where $g = 9.81\,\text{m/s}^2$ is the gravitational acceleration and $m$ is the maximum load mass.

Design safety factor:

$$S = 5.5 \tag{3}$$

Required critical (breaking) force:

$$F_{kr} = F_{\max} \cdot S = 7360 \times 5.5 \approx 40500\,\text{N} = 40.5\,\text{kN} \tag{4}$$

Taking into account that the rope may be wound in more than one layer, rope TEK 13310 is selected, which has breaking force $F_t = 59.7$ kN. Verification:

$$F_{\max} = 7.36\,\text{kN} < [F] = \frac{F_t}{S} = \frac{59.7}{5.5} \approx 10.9\,\text{kN} \tag{5}$$

Rope diameter:

$$d = 10\,\text{mm} \tag{6}$$

The drum diameter is calculated as:

$$D = e \cdot d = 20 \times 10 = 200\,\text{mm} \tag{7}$$

where $e = 20$ is the ratio depending on work conditions (range 20–30), selected for moderate industrial duty.

From the standard diameter series, the value $D = 200$ mm is adopted.

Drum length:

$$L_{tr} = 400\,\text{mm} \tag{8}$$

With groove pitch $t \approx d = 10$ mm, the number of rope turns per layer is:

$$\frac{L_{tr}}{t} = \frac{400}{10} = 40 \tag{9}$$

This accommodates approximately 25 m of rope per layer, sufficient for the workshop application.

Table 2: Rope and drum parameters

| Parameter | Value | Note |
|---|---|---|
| $F_{\max}$ | 7360 N | $mg$ |
| Safety factor $S$ | 5.5 | Standard for lifting |
| Required $F_{kr}$ | 40.5 kN | $F_{\max} \times S$ |
| Rope selected | TEK 13310 | $F_t = 59.7$ kN |
| Rope diameter $d$ | 10 mm | From catalogue |
| Drum diameter $D$ | 200 mm | $e = 20$, standard series |
| Drum length $L_{tr}$ | 400 mm | 25 m rope per layer |

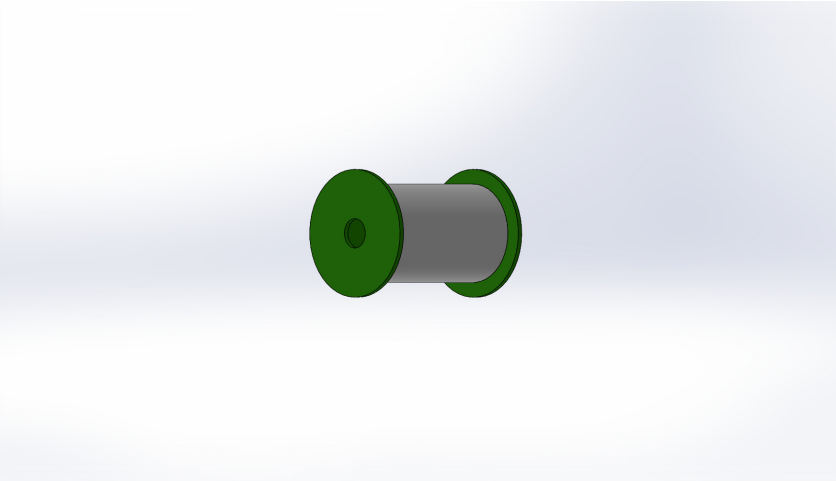


Figure 3: 3D model of the winch drum used for rope winding.

# 4 Gear Motor Calculation

## 4.1 Torque at the Drum

$$T = F_{\max} \cdot \frac{D}{2} = 7360 \times \frac{0.200}{2} = 736\,\text{N m} \tag{10}$$

## 4.2 Angular Velocity of the Drum

$$\omega_T = \frac{2v}{D} = \frac{2 \times 0.11}{0.200} = 1.1\,\text{rad/s} \tag{11}$$

## 4.3 Power Required at the Drum

$$P_T = T \cdot \omega_T = 736 \times 1.1 \approx 810\,\text{W} \tag{12}$$

## 4.4 Minimum Motor Power

Accounting for losses through all three transmission elements:

$$P_{M,\min} = \frac{P_T}{\eta_1 \eta_2 \eta_3} = \frac{810}{0.94 \times 0.92 \times 0.99} \approx 950\,\text{W} \tag{13}$$

where $\eta_1 = 0.94$ (spur gear motor), $\eta_2 = 0.92$ (chain drive), and $\eta_3 = 0.99$ (bearings).

Work power is taken as:

$$P_M \approx 1.2 \times P_{M,\min} \approx 1.14\,\text{kW} \tag{14}$$

## 4.5 Drum Rotation Frequency

$$n_T = \frac{30\omega_T}{\pi} = \frac{30 \times 1.1}{3.14} \approx 10.5\,\text{min}^{-1} \tag{15}$$

### 4.6 Chain Drive Ratio and Motor Speed

Chain drive ratio $u_K = 1.6$ is selected (range 1–7), giving:

$$n_R = n_T \times u_K = 10.5 \times 1.6 \approx 16.8\,\text{min}^{-1} \tag{16}$$

### 4.7 Motor Selection

Based on $P_M \geq 1.14$ kW and $n_R \approx 17\,\text{min}^{-1}$, gear motor R 77 DT 90S4 (SEW-Eurodrive) is selected.

Table 3: Selected gear motor

| Parameter | Value |
|---|---|
| Power $P_M$ | 1.1 kW |
| Output torque $M$ | 615 N m |
| Output speed $n_R$ | 17 $\text{min}^{-1}$ |
| Gear ratio $u_R$ | 81.8 |

### 4.8 Verification

Real drum speed:

$$n_{T,real} = \frac{n_R}{u_K} = \frac{17}{1.6} = 10.625\,\text{min}^{-1} \tag{17}$$

Real rope velocity:

$$v_{real} = \frac{\pi \cdot n_{T,real} \cdot D}{60} = \frac{3.14 \times 10.625 \times 0.200}{60} \approx 0.111\,\text{m/s} \tag{18}$$

Velocity error:

$$\Delta v = \frac{0.111 - 0.11}{0.11} \times 100\% = 0.9\% < 3\% \tag{19}$$

Power check:

$$P_{real} = P_M \cdot \eta_1 \cdot \eta_2 \cdot \eta_3 = 1.1 \times 0.94 \times 0.92 \times 0.99 \approx 0.94\,\text{kW} > P_T = 0.81\,\text{kW} \tag{20}$$

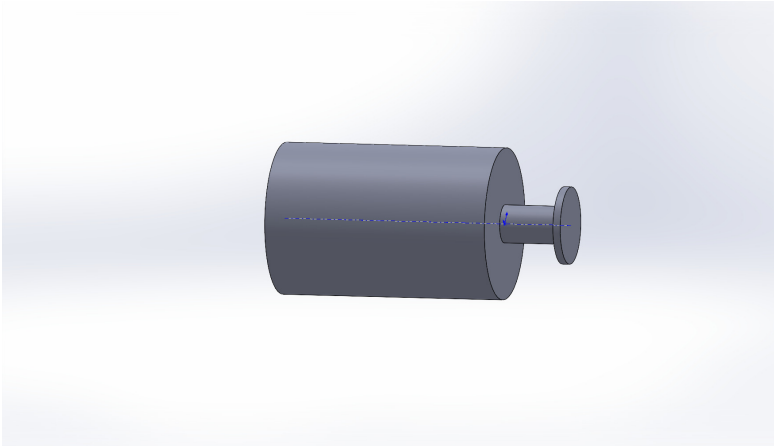


Figure 4: Selected gear motor model used to drive the system.

# 5 Calculation of Chain Drive

## 5.1 Torque on the Driving Sprocket

$$T_K = \frac{T}{u_K \eta_2 \eta_3} = \frac{736}{1.6 \times 0.92 \times 0.99} \approx 505\,\text{N m} \tag{21}$$

## 5.2 Number of Teeth

Recommended number of teeth on driving sprocket:

$$z_1 = 29 - 2u_K = 29 - 2 \times 1.6 = 26 \tag{22}$$

To minimise device size, $z_1 = 19$ is selected (low speed, minimum $z_{1,\min} = 13$–$15$).

$$z_2 = z_1 \times u_K = 19 \times 1.6 \approx 31 \tag{23}$$

## 5.3 Load Coefficient

$$K = k_d \times k_a \times k_n \times k_r \times k_m = 1.25 \times 1.0 \times 1.0 \times 1.25 \times 1.3 = 2.03 \tag{24}$$

where:

- $k_d = 1.25$ – variable/shock load,
- $k_a = 1.0$ – standard centre distance,
- $k_n = 1.0$ – inclination $\leq 60°$,
- $k_r = 1.25$ – manual periodic tensioning,
- $k_m = 1.3$ – periodic grease lubrication.

## 5.4 Chain Selection

Required pitch for single-strand chain: $t \geq 30$ mm (chain 20B1W, $t = 31.75$ mm). To minimise device size, a double-strand chain with smaller pitch is selected instead:

Selected: double-strand chain 16B2W (DIN 8187).

Table 4: Selected chain parameters

| Property | Value | Note |
|---|---|---|
| Pitch $t$ | 25.4 mm | |
| Mass $q$ | 4.21 kg/m | |
| Critical load $F_{kr}$ | 106 kN | |
| Roller width $b$ | 17.02 mm | |
| Roller diameter $d$ | 15.88 mm | |

### 5.5 Verification

Chain velocity:

$$v = \frac{z_1 t n_R}{60 \times 10^3} = \frac{19 \times 25.4 \times 17}{60000} \approx 0.14\,\text{m/s} \tag{25}$$

Circular force:

$$F_R = \frac{T_K \pi n_R}{30 v} = \frac{505 \times 3.14 \times 17}{30 \times 0.14} \approx 6400\,\text{N} \tag{26}$$

Roller pressure (double-strand, allowable pressure reduced 15%: $[p] = 47 \times 0.85 = 40$ MPa):

$$p = \frac{K F_R}{2bd} = \frac{2.03 \times 6400}{2 \times 17.02 \times 15.88} \approx 24\,\text{MPa} < [p] = 40\,\text{MPa} \tag{27}$$

Chain forces (centre distance $a = 800$ mm, $k_f = 1.5$ at 45°):

$$F_f = 9.81 \times k_f \times q \times a = 9.81 \times 1.5 \times 4.21 \times 0.8 \approx 50\,\text{N} \tag{28}$$

$$F_t = q v^2 = 4.21 \times 0.14^2 \approx 0.1\,\text{N} \tag{29}$$

Total shaft load:

$$F_V = F_R + 2F_f = 6400 + 2 \times 50 = 6500\,\text{N} \tag{30}$$

Safety factor of chain:

$$S = \frac{F_{kr}}{F_V + F_t + F_f} = \frac{106000}{6500 + 0.1 + 50} \approx 16 > [S] = 7.3 \tag{31}$$

### 5.6 Sprockets Selected

Table 5: Sprockets selected

| | Driving sprocket | Driven sprocket |
|---|---|---|
| Code | N16B2-19 | N16B2-31 |
| Teeth $z$ | 19 | 31 |
| Pitch diameter $D_0$ | 154.32 mm | 251.08 mm |
| Tip diameter $D_y$ | 165.2 mm | 262.0 mm |
| Bore $d$ | 20 mm | 25 mm |
| Width $L_n$ | 70 mm | 70 mm |
| Mass | 7.7 kg | 19 kg |

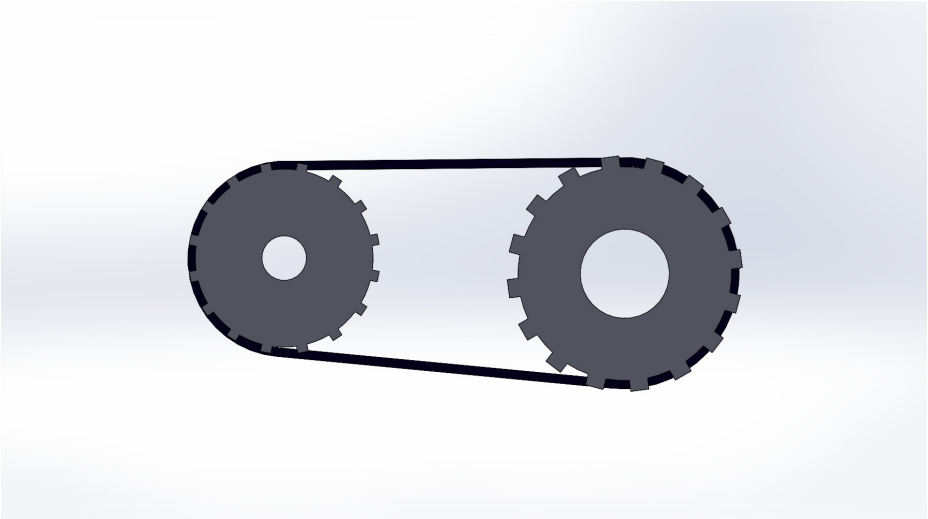


Figure 5: Chain drive system connecting the driving and driven sprockets.

# 6 Shaft Design

## 6.1 Minimum Shaft Diameter

From torsion using allowable shear stress $[\tau] = 30$ MPa:

$$d_v \geq \sqrt[3]{\frac{16T}{\pi[\tau]}} = \sqrt[3]{\frac{16 \times 736}{3.14 \times 30 \times 10^6}} \approx 0.050\,\text{m} \tag{32}$$

Selected: $d_v = 50$ mm. Diameter under bearing $d_t = 55$ mm. Internal hub diameter $d_r = 60$ mm. External hub diameter $d_{r2} = 100$ mm.

## 6.2 Hub Dimensions

$$l_r = 1.5 \times d_r = 1.5 \times 60 = 90\,\text{mm} \tag{33}$$

## 6.3 Shaft Layout

Table 6: Shaft layout dimensions

| Dimension | Value | Source |
|---|---|---|
| Chain wheel width $L_n$ | 70 mm | Sprocket catalogue |
| Bearing unit width $L_b$ | 60 mm | SY 55 TF |
| Hub length $l_r$ | 90 mm | Calculated |
| Drum length $L_{tr}$ | 400 mm | Chapter 3 |
| Bearing span $l$ | 520 mm | $L_{tr} + 2L_b$ |
| $l_1$ (C to A) | 75 mm | $L_n/2 + L_b/2 + 10$ |
| $l_2$ (A to D) | 90 mm | $l_r/2 + L_b/2 + 15$ |
| $l_3$ (D to E) | 340 mm | $l - 2l_2$ |

### 6.4 Reaction Forces

Taking moments about A and B:

$$R_B = \frac{\frac{F}{2}(l_2 + l_3) + \frac{F}{2}l_2 - F_V l_1}{l} = 2740\,\text{N} \approx 2.74\,\text{kN} \tag{34}$$

$$R_A = \frac{\frac{F}{2}(l - l_2 - l_3) + \frac{F}{2}(l - l_2) + F_V(l + l_1)}{l} = 11100\,\text{N} \approx 11.1\,\text{kN} \tag{35}$$

Check:

$$R_A + R_B = 11100 + 2740 = 13840 \approx F_V + F = 6500 + 7360 = 13860\,\text{N} \tag{36}$$

### 6.5 Bending Moments

$$M_A = F_V l_1 = 6500 \times 0.075 = 488\,\text{N m} \quad \text{(critical section)} \tag{37}$$

$$M_D = F_V(l_1 + l_2) - R_A l_2 = 74\,\text{N m} \tag{38}$$

$$M_E = R_B(l - l_2 - l_3) = 243\,\text{N m} \tag{39}$$

Torque $T = 736$ N m acts from C through D (full value), then reduces to 368 N m from D to E (split between two hubs).

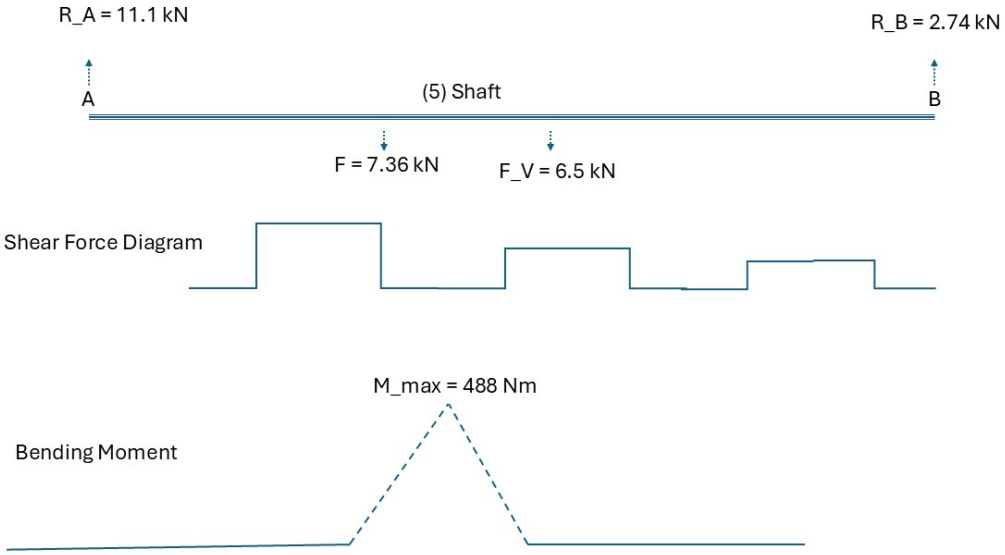



Figure 6: Internal force diagram of the shaft showing reaction forces, shear forces, and bending moment distribution.

### 6.6 Equivalent Moment and Stress Check

$$M_{ekv} = \sqrt{M_A^2 + 0.75T^2} = \sqrt{488^2 + 0.75 \times 736^2} \approx 803\,\text{N m} \tag{40}$$

$$\sigma_{ekv} = \frac{32M_{ekv}}{\pi d_t^3} = \frac{32 \times 803}{3.14 \times 0.055^3} \approx 49\,\text{MPa} \tag{41}$$

$$[\sigma] = \frac{R_{p0.2}}{S} = \frac{370}{1.5} \approx 247\,\text{MPa} \tag{42}$$

$$49\,\text{MPa} < 247\,\text{MPa} \tag{43}$$

### 6.7 Fatigue Safety Factor (Stress Concentration Check)

Stress concentration at bearing A (diameter step $d_t/d_v = 55/50$, fillet $R = 1$ mm). For C45E ($R_m = 630$ MPa, $d = 55$ mm):

$$K_\sigma = 1.96, \quad K_\tau = 1.30, \quad K_{d\sigma} = 0.83, \quad K_{d\tau} = 0.69, \quad K_F = 0.95, \quad \psi_\sigma = 0.2, \quad \psi_\tau = 0.1 \tag{44}$$

Amplitude bending stress:

$$\sigma_a = \frac{32M_A}{\pi d_t^3} = \frac{32 \times 488}{3.14 \times 0.055^3} \approx 29.9\,\text{MPa}, \quad \sigma_m = 0 \tag{45}$$

Safety factor for bending:

$$S_\sigma = \frac{\sigma_{-1}}{\frac{K_\sigma}{K_F K_{d\sigma}}\sigma_a} = \frac{275}{\left(\frac{1.96}{0.95 \times 0.83}\right) \times 29.9} \approx 3.7 \tag{46}$$

Amplitude torsion stress:

$$\tau_a = \tau_m = \frac{16T}{2\pi d_t^3} = \frac{16 \times 736}{2 \times 3.14 \times 0.055^3} \approx 11.3\,\text{MPa} \tag{47}$$

Safety factor for torsion:

$$S_\tau = \frac{\tau_{-1}}{\frac{K_\tau}{K_F K_{d\tau}}\tau_a + \psi_\tau \tau_m} = \frac{165}{22.4 + 1.1} \approx 7.0 \tag{48}$$

Combined safety factor:

$$S = \frac{S_\sigma S_\tau}{\sqrt{S_\sigma^2 + S_\tau^2}} = \frac{3.7 \times 7.0}{\sqrt{3.7^2 + 7.0^2}} \approx 3.3 \tag{49}$$

$$S = 3.3 > [S] = 2.5 \ldots 3.0 \tag{50}$$

The designed shaft diameters are accepted.

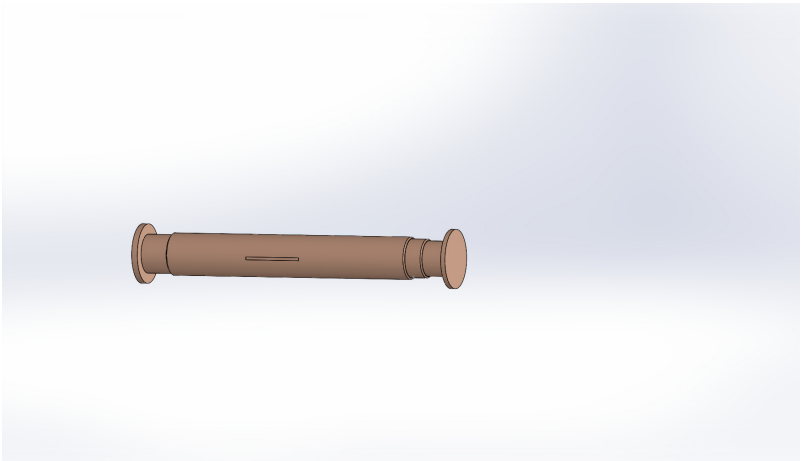


Figure 7: 3D model of the designed shaft.

# 7 Bearings

Self-aligning bearing units are required per the initial conditions. SKF SY plummer block units with self-aligning ball bearings are selected. For shaft diameter $d_t = 55$ mm, the selected unit is SY 55 TF.

Table 7: Bearing selection

| Component | Designation |
|---|---|
| Complete unit | SY 55 TF |
| Bearing insert | YAR 211-2F |
| Housing | SY 511 M |

Table 8: Bearing parameters

| Parameter | Value |
|---|---|
| Shaft diameter $d$ | 55 mm |
| Dynamic load rating $C$ | 43.6 kN |
| Static load rating $C_0$ | 29 kN |
| Limiting speed | 3600 $\text{min}^{-1}$ |
| Mass | 3.6 kg |

## 7.1 Bearing Life Calculation

Calculated for bearing A (maximum load $R_A = 11100$ N):

$$L_{10} = \left(\frac{C}{F_A}\right)^3 \frac{10^6}{60n} = \left(\frac{43600}{11100}\right)^3 \frac{10^6}{60 \times 10.5} \approx 96100\,\text{h} \tag{51}$$

Minimum required bearing life for industrial lifting: $L_{10,\text{min}} = 14000$ h.

$$96100\,\text{h} \gg 14000\,\text{h} \tag{52}$$

Bearing B verification:

$$L_{10,B} \approx 6390000\,\text{h} \tag{53}$$

The bearing unit width $L_b \approx 60$ mm used in the shaft layout is confirmed by the SY 55 TF catalogue dimensions.

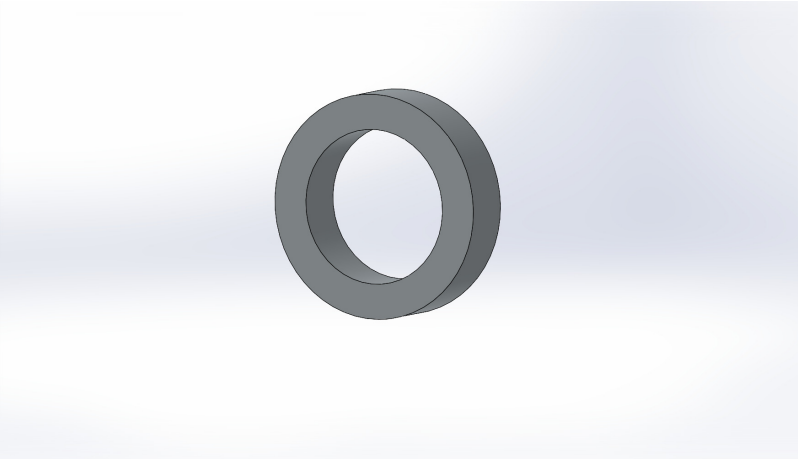


Figure 8: Self-aligning bearing unit used to support the shaft.

# 8 Key Calculation

Keys are selected from Annex 1, Table 5. The same key cross-section 14×9 mm is used at both shaft locations to simplify manufacturing (single keyway cutter for both diameters).

Table 9: Key dimensions

| Parameter | Shaft–drum joint | Shaft–chain wheel joint |
|---|---|---|
| Location diameter | $d_r = 60$ mm | $d_v = 50$ mm |
| Key cross-section ($b \times h$) | 14×9 mm | 14×9 mm |
| $t_1/t_2$ | 5.5 / 3.8 mm | 5.5 / 3.8 mm |
| Hub/wheel width | $l_r = 90$ mm | $L_n = 70$ mm |
| Key length $l_{key}$ | 80 mm | 63 mm |
| Number of keys | 2 (one per hub) | 2 (180° apart) |

## 8.1 Joint 1: Shaft–Drum (at $d_r = 60$ mm, two keys)

Contact (bearing) stress with two keys:

$$\sigma_C = \frac{2T}{2d_r(h - t_1)(l_{key} - b)} = \frac{1472}{2 \times 0.060 \times 0.0035 \times 0.066} \approx 53\,\text{MPa} < [\sigma]_C = 130\,\text{MPa} \tag{54}$$

Shear stress:

$$\tau = \frac{2T}{2d_r b(l_{key} - b)} = \frac{1472}{2 \times 0.060 \times 0.014 \times 0.066} \approx 13\,\text{MPa} < [\tau] = 70\,\text{MPa} \tag{55}$$

## 8.2 Joint 2: Shaft–Chain Wheel (at $d_v = 50$ mm, two keys)

Single key contact stress = 172 MPa > 130 MPa, so single key fails. Two keys are used (180° apart):

$$\sigma_C = \frac{2T}{2d_v(h - t_1)(l_{key} - b)} = \frac{1472}{2 \times 0.050 \times 0.0035 \times 0.049} \approx 86\,\text{MPa} < [\sigma]_C = 130\,\text{MPa} \tag{56}$$

Shear stress:

$$\tau = \frac{2T}{2d_v b(l_{key} - b)} = \frac{1472}{2 \times 0.050 \times 0.014 \times 0.049} \approx 21\,\text{MPa} < [\tau] = 70\,\text{MPa} \tag{57}$$

All strength conditions are satisfied at both joints.

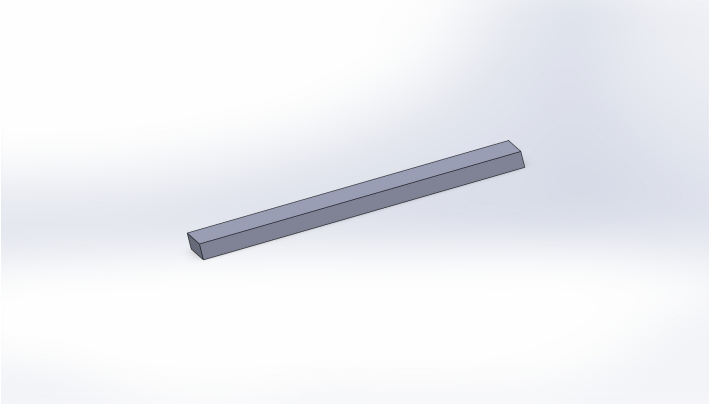


Figure 9: Key used for torque transmission between shaft and connected components.

# 9 Drum Calculation

Drum material: S235J2G3 EN 10025. Design stress $[\sigma] = 120$ MPa. Rope diameter $d = 10$ mm.

## 9.1 Wall Thickness

Groove pitch $t \approx d = 10$ mm. Minimum wall thickness:

$$\delta \geq \frac{F_{\max}}{t[\sigma]} = \frac{7360}{0.010 \times 120 \times 10^6} = 0.00061\,\text{m} = 0.61\,\text{mm} \tag{58}$$

Selected: $\delta = 7$ mm (standard plate, groove depth $b \approx \delta$).

$$D_1 = D - 2\delta = 200 - 14 = 186\,\text{mm} \tag{59}$$

## 9.2 Drum as a Beam

The drum shell is modelled as a simply supported beam of span $l_3 = 340$ mm loaded by $F = 7360$ N at midspan. By symmetry:

$$R_D = R_E = \frac{F}{2} = 3680\,\text{N} \tag{60}$$

$$M = R_D \times \frac{l_3}{2} = 3680 \times 0.170 \approx 626\,\text{N m} \tag{61}$$

## 9.3 Section Moduli of Drum Shell

$$W = 0.1 \times \frac{D^4 - D_1^4}{D} = 0.1 \times \frac{0.200^4 - 0.186^4}{0.200} \approx 2.01 \times 10^{-4}\,\text{m}^3 \tag{62}$$

$$W_0 = 2W = 4.02 \times 10^{-4}\,\text{m}^3 \tag{63}$$

### 9.4 Stress Components

$$\sigma_S = \frac{F_{\max}}{\delta t} = \frac{7360}{0.007 \times 0.010} = 105\,\text{MPa} \tag{64}$$

$$\sigma_M = \frac{M}{W} = \frac{626}{2.01 \times 10^{-4}} \approx 3.1\,\text{MPa} \tag{65}$$

$$\tau = \frac{T}{W_0} = \frac{736}{4.02 \times 10^{-4}} \approx 1.8\,\text{MPa} \tag{66}$$

### 9.5 Equivalent Stress

$$\sigma_{ekv} = \sqrt{(sigma_S + \sigma_M)^2 + 4\tau^2} = \sqrt{(105 + 3.1)^2 + 4 \times 1.8^2} \approx 108\,\text{MPa} \tag{67}$$

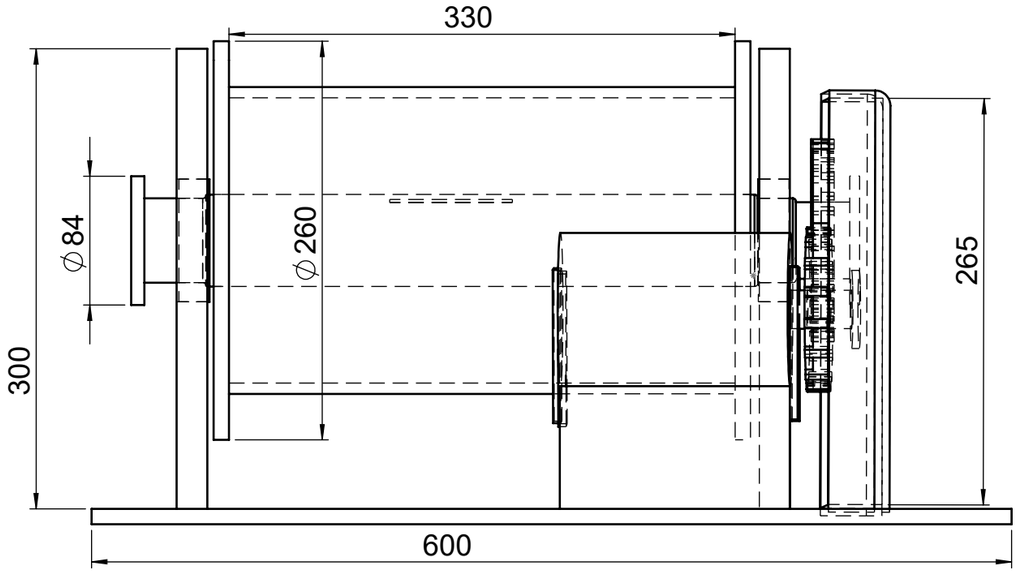
$$108\,\text{MPa} < [\sigma] = 120\,\text{MPa} \tag{68}$$

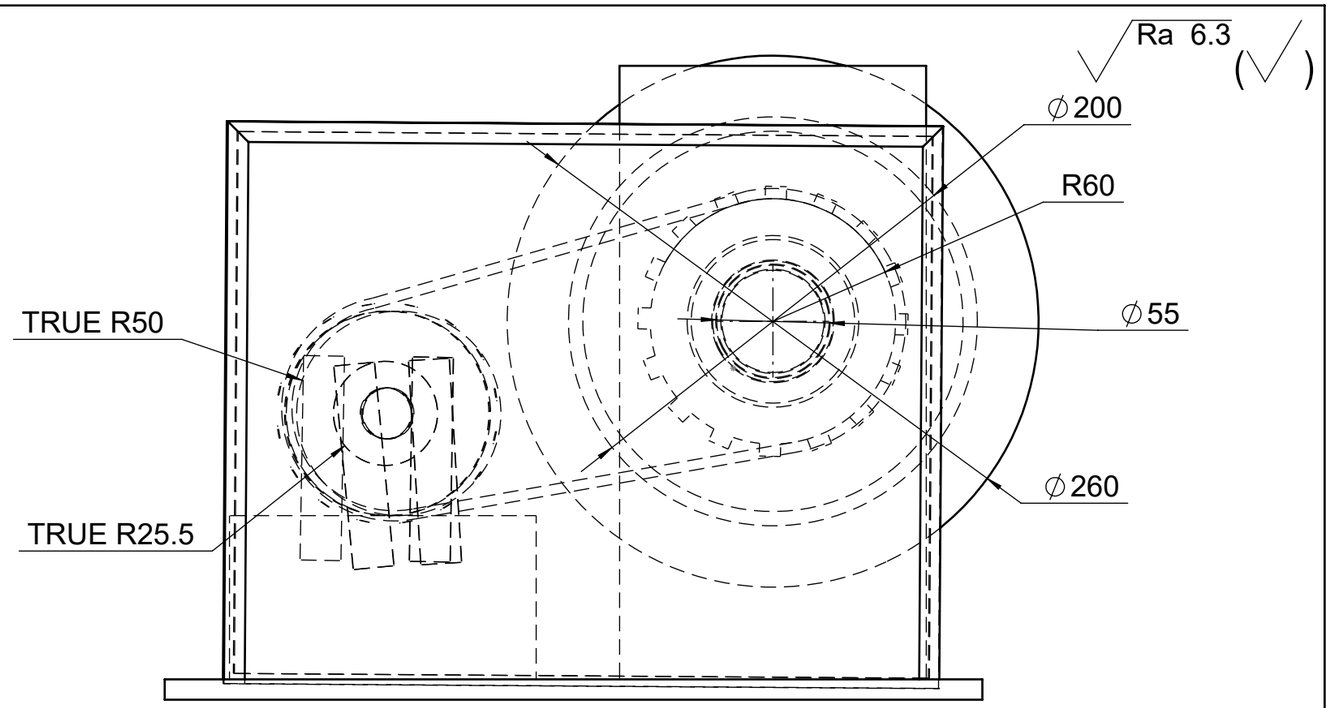
The drum wall thickness of 7 mm is accepted. Wall utilisation is 90%, confirming the selection is efficient.

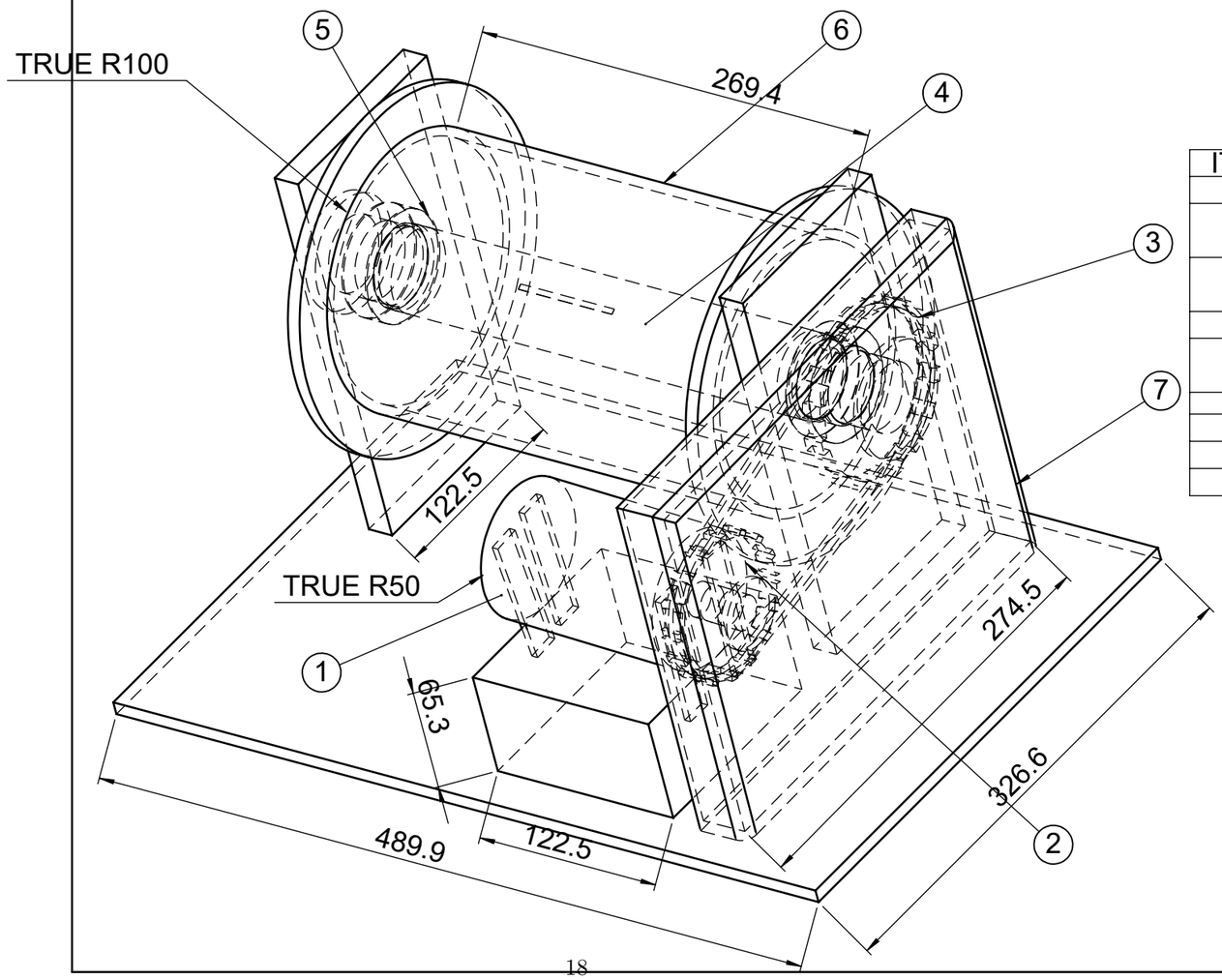
Table 10: Drum parameters

| Parameter | Value |
|---|---|
| Wall thickness $\delta$ | 7 mm |
| Inner diameter $D_1$ | 186 mm |
| Drum span $l_3$ | 340 mm |
| Max bending moment $M$ | 626 N m |
| Compressive stress $\sigma_S$ | 105 MPa |
| Bending stress $\sigma_M$ | 3.1 MPa |
| Torsional stress $\tau$ | 1.8 MPa |
| Equivalent stress $\sigma_{ekv}$ | 108 MPa |
| Allowable stress $[\sigma]$ | 120 MPa |
| Result | $108 < 120$ MPa |

## Final Assembly Drawing




| ITEM NO. | | DESCRIPTION | QTY. | MATERIAL |
|---|---|---|---|---|
| 1 | Motor SEW R77 DT90S4 | | 1 | Cast aluminum / steel |
| 2 | Driving sprocket N16B2-19 | 19 teeth sprocket | 1 | Hardened steel |
| 3 | Driven sprocket N16B2-31 | 31 teeth sprocket | 1 | Hardened steel |
| 4 | | EW 01-02-00 | 1 | C45E steel |
| 5 | SKF SY 55 TF | Bearing | 2 | Standard SKF bearing steel |
| 6 | Drum | | 1 | S235 steel |
| 7 | Chain Cover + Stand | | 1 | Structural steel S355 |
| | Key | 14*9*80 DIN 6885 | 1 | Steel (C45 or standard) |
| | Frame + Support | | 1 | Structural steel S355 |

| | | | | |
|---|---|---|---|---|
| | Materjal: SEE BOM | Märkimata piirhälbed: ISO 2768-m; ISO 2768-K | Mass: 30-55kg | Mõõt: 1:10 |
| Teostas: | 03/06/2026 | Nimetus: Jordan Thomas Brodie<br>ELECTRIC WINCH DRAWING<br>ID 243483 | | |
| Kontrollis: | - | | | |
| Kinnitas: | - | | | |
| TAL TECH Tallinna Tehnikaülikool<br>Mehaanika ja tööstustehnika instituut<br>Ehitajate tee 5, Tallinn, 19086 | Leht: 1 / 1 | Formaat: A3 | Tähis: Winch | |



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# 10 References


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